

When citations lack support

A reproducible audit of published ALCE annotations, with an explicit data inventory and question-cluster uncertainty estimates.

2,604

annotated sentences

32.5%

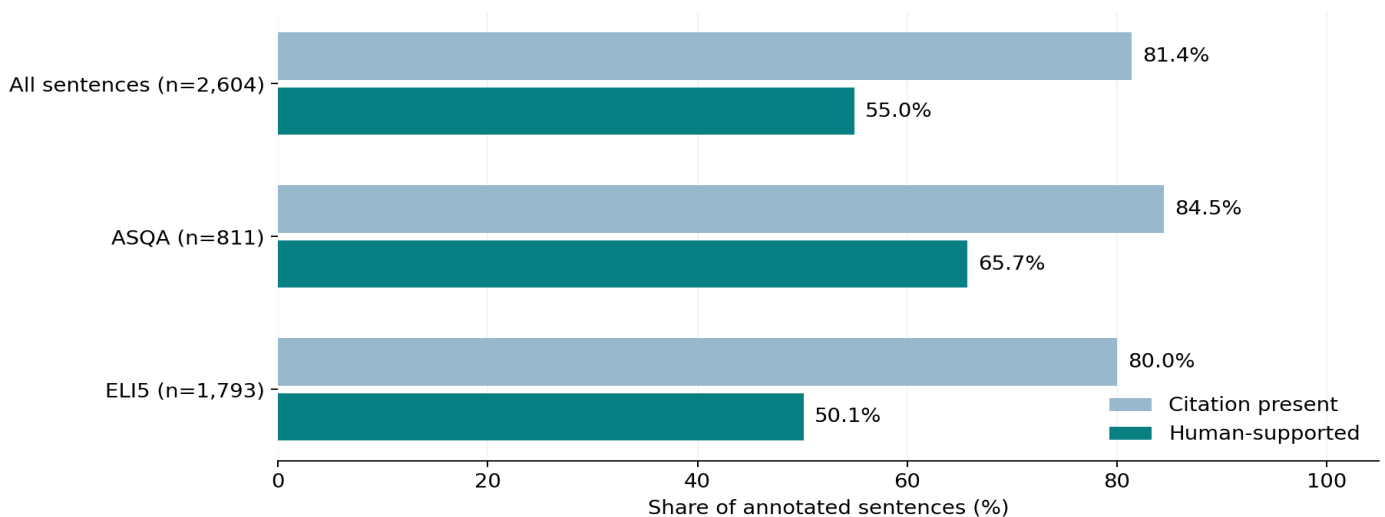
of cited sentences lack full support

14.9%

automatic / human disagreement

The finding

Of 2,120 cited sentences, 688 lack full support under the published human labels. The estimated share is 32.5% (95% question-cluster interval: 29.4%-35.6%). A citation count alone would miss this evidence gap.



What was measured

Measure	Count / denominator	Rate
Citation present	2,120 / 2,604 sentences	81.4%
Human-supported	1,432 / 2,604 sentences	55.0%
Automatic / human disagreement	388 / 2,604 sentences	14.9%

Reading the chart: sentence-weighted rates across eight historical configurations. Citation presence means a citation in the annotated list; support means support from supplied passages. It does not mean independently verified factual truth.

How the result was checked

A complete record of the inputs, denominators and limits of this small research study.

Data inventory and method

800 answer records across ASQA and ELI5: 100 shared questions per task, four configurations per task. 22 empty answers are retained in the inventory; the remaining 778 contribute 2,604 sentences. All 200 task/question pairs have at least one non-empty answer.

The analysis checks the source SHA256, validates sentence alignment and binary labels, then exports sentence, answer and configuration tables. It reuses the original human and automatic labels. Eight automated tests passed.

Intervals use 2,000 bootstrap samples with seed 20260922, resampling question clusters within each task while retaining configurations and sentences. They quantify question-sample variability, not annotation error or market-wide uncertainty.

Automatic assessment versus human labels

	Human supported	Human unsupported
Automatic supported	1,251	207
Automatic unsupported	181	965

The automatic and human aggregate support rates are close (56.0% versus 55.0%), yet 388 individual labels differ. Aggregate similarity can conceal item-level disagreement.

Interpretation and limits

Use separate measures for visibility, citation presence and evidence support. This benchmark audit is a completed secondary study; it does not measure current commercial engine rankings or demonstrate SEO uplift. Retrieval methods and document counts vary between configurations, so comparisons are descriptive. No new live-engine queries or human ratings were collected.

Sources and reproducibility

Gao, Yen, Yu & Chen (2023), Enabling Large Language Models to Generate Text with Citations, EMNLP. arxiv.org/abs/2305.14627

Source release: github.com/princeton-nlp/ALCE, commit 246c476a4edf. Upstream MIT notice retained. Full hash and source paths: SOURCE.md.

GEO context: Aggarwal et al. (2024), GEO: Generative Engine Optimization, KDD. arxiv.org/abs/2311.09735. The accompanying LIVE_EXPERIMENT_PROTOCOL.md proposes a separate 108-response English pilot; it has not been executed.